

Sowjanya Krishna

Research Scientist (Industrial IoT & Digital Twins) — M.Sc., B.Eng

📍 Stuttgart, Germany

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»»» Experience

Research Scientist, IT Systems - Fraunhofer IPA, Stuttgart

12/2023 - Present

- Designed and implemented industrial digitalization workflows connecting data sources, metadata catalogues, digital tools, and digital twin systems.
- Developed digital twin and data-space concepts for KI Data Platform using AAS, BaSyx/FA3ST, DCAT-AP, and Manufacturing-X/Catena-X principles.
- Built time-series abstraction workflows for H2Giga using InfluxDB, MQTT/event triggers, BaSyx, and IDTA Time Series Submodels to standardize machine and sensor data.
- Implemented IT/OT data integration for ExELPro using SmartUnifier, OPC UA, InfluxDB, edge devices, and ML-ready production datasets.
- Connected production, energy, and sustainability data for DigiBattPro 4.0 with LCA/PCF workflows, dashboards, and interoperable digital twin representations.
- Architected a digital twin concept for Fraunhofer FFB Lab Hub, linking lab rooms, device booking, and lab data systems into one operational platform.
- Designed shop-floor dashboards and digital-twin UIs to support hotspot detection, downtime analysis, process monitoring, and sustainability decisions from sensor and process data.

Working Student, Smart Building IoT - ARENA 2036 & SPIE, Stuttgart

04/2023 - 08/2023

- Developed end-to-end IoT data workflows for smart building and industrial monitoring, enabling sensor data to be collected, processed, stored, and visualized for operational insights.
- Built Azure-based data handling workflows to collect, process, and structure sensor data for scalable analytics.
- Created Grafana dashboards for real-time visualization of environmental conditions, sensor trends, and industrial monitoring data.
- Integrated 5G/IIoT sensor data into monitoring views, supporting smart building automation and data-driven operations.

Intern, IoT Edge Software - Robert Bosch Power Tools GmbH, Stuttgart

05/2022 - 10/2022

- Implemented over-the-air software rollout for IoT edge devices, enabling remote updates via Bosch IoT Suite.
- Developed standardized MQTT-based edge-to-cloud communication workflows for software-updatable IoT devices.
- Built software-updatable device feature models using Eclipse Vorto and Hawkbit to support standardized IoT device management.

Technical Skills

Smart Building & Industrial IoT	MQTT, OPC UA, 5G/IIoT, Raspberry Pi, InfluxDB
IT/OT Integration & Data Pipelines	SmartUnifier, Edge-PC, Xetics, Telegraf, ETL/ELT workflows
Digital Twin & Industrial Standards	Asset Administration Shell, BaSyx, FA3ST, IDTA
Backend & APIs	Python, REST APIs, JSON, Javascript, InfluxDB, MongoDB
Frontend	Grafana, Power BI, Vue.js, React, PyQt, Next.js, Dash
Sustainability & ESG Data	PCF, LCA, LCIA, OpenLCA, Umberto, KNIME
Embedded & Edge Software	Bosch IoT Suite, Bosch IoT Rollouts, Eclipse Vorto, Hawkbit, Linux
Cloud & Data Platforms	Microsoft Azure, Piveau, DCAT-AP, Docker, Keycloak, Eclipse Dataspace Connector, Manufacturing-X

Education

M.Sc. Information Technology - University of Stuttgart, Germany	10/2020 - 08/2023
» Industrial Automation, Real-time Concepts, Deep Learning, Smart Cities and IoT	
B.Eng. Electronics and Communication - Alliance University, Bengaluru	07/2015 - 08/2019
» Embedded Systems, Communication Networks, Control Systems, Industrial Electronics	

Projects

Master Thesis — Human Detection for Autonomous Mobile Robots - ARENA2036	02/2023 - 08/2023
» Developed a real-time human detection pipeline using LIDAR occupancy maps, U-Net, YOLOv5, and camera/LIDAR sensor fusion.	
» Improved indoor logistics safety by enabling human localization for autonomous mobile robot operation.	
Research Project — Dynamic Geofence for Autonomous Mobile Robots - ARENA2036	12/2021 - 05/2022
» Built a real-time AMR safety application using Vue.js, Flask, MQTT, SQLite, and computational geometry.	
» Developed dynamic risk-zone visualization to support safer robot navigation in indoor logistics environments.	
University Project — Smart Farming using HVAC and IoT - University of Stuttgart	04/2021 - 07/2021
» Developed an IoT automation prototype using Raspberry Pi, MQTT, sensor integration, and actuator-control logic.	
» Built a Dash-based monitoring interface and PDDL planner for automated decision-making and HVAC-related control.	

Publications

- » An Automated Workflow for the PCF Calculation of Battery Cells. May 2026. DOI: 10.1007/978-3-032-21157-6_22
- » Leveraging Digital Twins for Real-Time Environmental Monitoring in Battery Manufacturing. <https://doi.org/10.1016/j.procir.2024.10.159>
- » CCPT Conference - Sovereign Data Spaces and Interoperable Digital Twins for Automated PCF (Yet to be published.)

Languages

- » **English:** C1 (Professional Proficiency)
- » **German:** B1 (Intermediate Proficiency)